

Agenda:

1. Sum of all elements from L to R
 2. Prefix Sum
 3. Sum of even indexed elements
 4. Special Index
 5. Maximum subarray sum with length K
 6. Find sum of all subarrays sums
-

Q → Given an integer array of size N & a queries, for every query → find sum from L to R.

0 1 2 3 4 5 6
A = [-3 6 2 4 -5 2 8]

L	R
1	3 → 12
2	6 → 11
0	4 → 4

$$1 \leq N \leq 10^5$$

$$1 \leq Q \leq 10^5$$

Bruteforce → $\forall$ queries, iterate from L to R & get sum.

```
for i → 0 to (Q-1) { // B[Q][2]
```

```
    L = B[i][0]    R = B[i][1]
```

```
    sum = 0
```

```
    for j → L to R {
```

```
        sum += A[j]
```

```
    }
```

```
    print (sum)
```

```
}
```

TC = $O(Q \times N)$ SC = $O(1)$

$\sim 10^5 \times 10^5 \rightarrow 10^{10}$ iterations

$\Rightarrow$ TLE



OVERS	1	2	3	4	5	6	7	8	9	10
SCORE	2	8	14	29	31	49	65	79	88	97

Scoreboard after every over.

Prefix sum

Runs scored in 7th over $\rightarrow 65 - 49 = \underline{16}$

Runs scored from 6th to 10th over $\rightarrow 97 - 31 = \underline{66}$

Runs scored in 10th over $\rightarrow 97 - 88 = \underline{9}$

Prefix Sum

$$P[i] = A[0] + A[1] + \dots + A[i]$$

$$P[3] = \underbrace{A[0] + A[1] + A[2]}_{P[2]} + A[3]$$

$$P[2]$$

$$P[i] = P[i-1] + A[i], \quad i > 0$$

$$P[0] = A[0]$$

$$A = \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ [10 & 32 & 6 & 12 & 20 & 1] \end{matrix}$$

$$P = [10 \quad 42 \quad 48 \quad 60 \quad 80 \quad 81]$$

$$P[0] = A[0]$$

```
for i → 1 to (N-1) {  
    P[i] = P[i-1] + A[i]  
}
```

$$x + y - y = x$$

$$TC = \underline{O(N)}$$

$$\text{Sum } L \text{ --- } R \rightarrow \boxed{A[L] + A[L+1] + \dots + A[R] + (A[0] + A[1] + \dots + A[L-1])} \rightarrow P[R]$$

$$- \boxed{(A[0] + A[1] + \dots + A[L-1])} \rightarrow P[L-1]$$

$$\boxed{\text{Sum } L \text{ --- } R = P[R] - P[L-1]}$$

```

for i → 0 to (Q-1) { // B[Q][2]
    L = B[i][0]    R = B[i][1]
    if (L == 0) sum = P[R]
    else sum = P[R] - P[L-1]
    print (sum)
}

```

TC = $O(Q)$

Overall → TC = $O(N + Q)$

SC = $O(N)$ → P[]

$O(1)$

~~A[0] = A[0]~~

```

for i → 1 to (N-1) {
    A[i] = A[i-1] + A[i]
}

```

$\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \\ A = [10 & \cancel{32} & \cancel{6} & \cancel{12} & \cancel{20} & \cancel{1}] \end{matrix}$

↳ [10 42 48 60 80 81]

Q → Given an integer array of size N & Q queries,
for every query → find sum from L to R of
only even index elements.

$$A = \begin{matrix} & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ [-3 & 6 & 2 & 4 & -5 & 2 & 8] \end{matrix}$$

<u>L</u>	<u>R</u>
1	3 → 2
2	6 → 5
0	4 → -6

$$1 \leq N \leq 10^5$$

$$1 \leq Q \leq 10^5$$

$$P[0] = A[0]$$

for $i \rightarrow 1$ to $(N-1)$ {

if $(i \% 2 == 0)$ $P[i] = P[i-1] + A[i]$ ←

else $P[i] = P[i-1]$ ←

}

$$A = \begin{matrix} & 0 & 1 & 2 & 3 & 4 \\ [2 & 4 & 3 & 1 & 5] \end{matrix}$$

$$P = [2 \quad 2 \quad 5 \quad 5 \quad 10]$$

```

for i → 0 to (Q-1) { // B[A][2]
    L = B[i][0]    R = B[i][1]
    if (L == 0) sum = P[R]
    else sum = P[R] - P[L-1]
    print (sum)
}

```

Overall → TC = $O(N + Q)$

SC = $O(N)$ → P[]

use A[] for storing
prefix sum → $O(1)$

Q → Given an integer array, count the no. of special index.

Index after removing which,
sum of even index elements = sum of odd index elements.

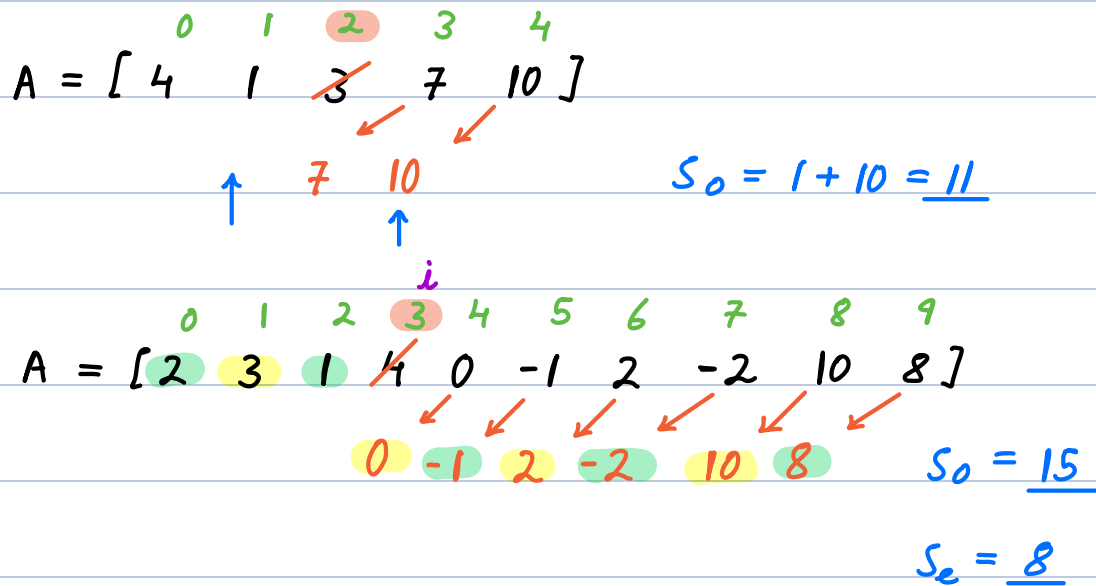
	0	1	2	3	4	5		
i	A = [4 3 2 7 6 -2]						Se	So
0	3	2	7	6	-2		8	8 ✓
1	4	2	7	6	-2		9	8
2	4	3	7	6	-2		9	9 ✓
3	4	3	2	6	-2		4	9
4	4	3	2	7	-2		4	10
5	4	3	2	7	6		12	10

Ans = 2

Bruteforce $\rightarrow$ $\forall i$, find S_e & S_o after removal of $A[i]$.

If $S_e == S_o$ increase answer by 1.

$$TC = \underline{O(N^2)} \quad SC = \underline{O(1)}$$



if i^{th} element is removed $\rightarrow$

$S_e =$ Sum of even index from 0 to $(i-1)$ +
Sum of odd index from $(i+1)$ to $(N-1)$

$$S_e = P_e[i-1] + (P_o[N-1] - P[i])$$

$$S_o = P_o[i-1] + (P_e[N-1] - P_e[i])$$

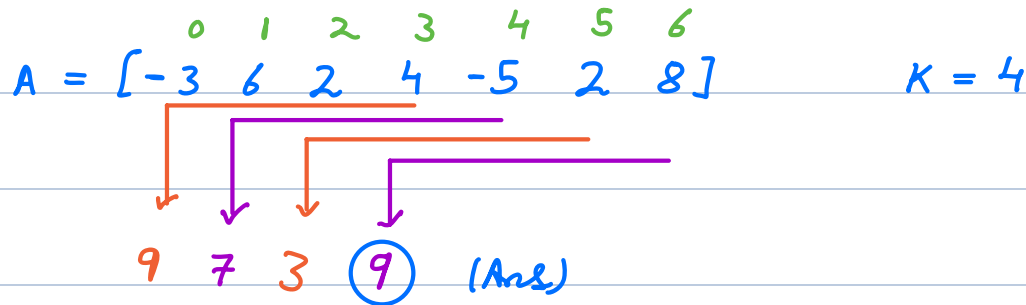
Sol $\rightarrow$ 1) Calculate P_e & P_o arrays

2) $\forall i$, calculate S_e & S_o

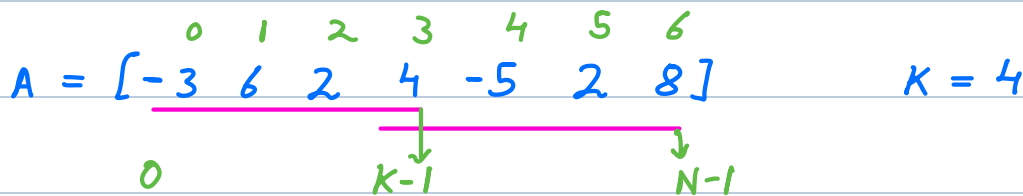
3) if $(S_e == S_o)$ ans++

$$TC = \underline{O(N)} \quad SC = \underline{O(N)}$$

Q → Given an integer array, find max subarray sum, for subarray of $len = K$.



Bruteforce → $\forall$ subarrays of $len = K$, find sum & store max as answer.



$$\text{end} \rightarrow [(K-1) \quad (N-1)] \rightarrow N-1 - K+1 + 1$$

$$= \underline{\underline{N-K+1}}$$

$$TC = \underline{O((N-K+1) * K)}$$

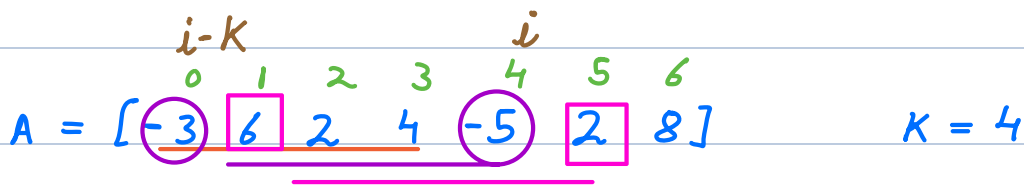
$$K = N \rightarrow (N-N+1) * N = \underline{\underline{N}}$$

$$K = 1 \rightarrow (N-1+1) * 1 = \underline{\underline{N}}$$

$$K = \frac{N}{2} \rightarrow (N - \frac{N}{2} + 1) * \frac{N}{2} = \frac{N}{2} * \frac{N}{2} = \frac{N^2}{4}$$

$$TC = O(N^2)$$

Sliding window



sum = 13

sum = 0

```
for i → 0 to (K-1) {  
    sum += A[i]  
}
```

ans = sum

```
for i → K to (N-1) {  
    sum += A[i] - A[i-K]  
    ans = max(ans, sum)  
}
```

return ans

TC = $O(N)$ SC = $O(1)$

Q → Given an integer array, find sum of all possible subarray sums.

Sum

$$A = [\overset{0}{3} \overset{1}{2} \overset{2}{5}]$$

$$3 \rightarrow 3$$

$$3 \ 2 \rightarrow 5$$

$$3 \ 2 \ 5 \rightarrow 10$$

$$2 \rightarrow 2$$

$$2 \ 5 \rightarrow 7$$

$$5 \rightarrow + \underline{5}$$

$$\underline{32} \text{ (Ans)}$$

Brute force $\rightarrow$ $\forall$ subarrays

calculate sum &

add in the answer.

$$TC = O(N^2 * N) \rightarrow O(N^3)$$

Prefix sum $\rightarrow TC = O(N^2)$

$$SC = O(N) / O(1)$$

$$TC = O(N)$$

Contribution Technique

$$Ans = \sum_{i} \text{contribution of } A[i]$$

$$A[i] * (\# \text{ subarrays where } A[i] \text{ is present})$$

HW

$$A = [\overset{0}{3} \overset{1}{-2} \overset{2}{4} \overset{3}{-1} \overset{4}{2} \overset{5}{6}]$$

$$\text{start} \rightarrow [0 \ 1] \rightarrow 2$$

$$\text{end} \rightarrow [1 \ 5] \rightarrow 5$$

$$> 2 * 5 = \underline{10}$$